# AQ11.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.

### Level 1

*1 point*

Choose the correct statements about $f(x, y) = x^3 + xy + y^4 - 4$.

- [ ] $(0, 0)$ is a critical point of $f$.
- [ ] $(0, 0)$ is the only critical point of $f$ in $R^2$.
- [ ] $f_{xx} = 6x$
- [ ] $f_{yy} = 12x^2$
- [ ] $f_{xy} = 1$
- [ ] $f_{yx} = 0$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$(0, 0)$ is a critical point of $f$.

$f_{xx} = 6x$

$f_{xy} = 1$

*1 point*

If $f(x, y) = \tan^{-1}(x/y)$, then which of the following options are correct.

- [ ] $f_x = \frac{y}{(x^2 + y^2)}$
- [ ] $f_{yy} = \frac{2xy}{(x^2 + y^2)^2}$
- [ ] $f_{yy} = \frac{-2xy}{(x^2 + y^2)^2}$
- [ ] $f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$
- [ ] $f_{yx} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$f_x = \frac{y}{(x^2 + y^2)}$

$f_{yy} = \frac{2xy}{(x^2 + y^2)^2}$

$f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$

*1 point*

If $f(x, y) = x^2 - 3xy^3$ f(x, y) = x2 − 3xy3, then the determinant of the Hessian matrix $Hf(x, y)$ Hf(x, y) is

- ○ $36xy + 81y^4$ 36xy + 81y4
- ○ $0$ 0
- ○ $-36xy - 81y^4$ −36xy − 81y4
- ○ $36xy - 81y^4$ 36xy − 81y4

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$-36xy - 81y^4$ −36xy − 81y4

**Level 2**

*1 point*

Choose the correct statements about $f(x, y) = x^2 + y^4$.f(x, y) = x2 + y4.

- ☐ $(0, 0)$(0, 0) is a critical point of $f$f.
- ☐ Rank of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is $2$2.
- ☐ The determinant of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is $0$0.
- ☐ The determinant of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is $1$1.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$(0, 0)$(0, 0) is a critical point of $f$f.
The determinant of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is $0$0.

Number of critical points of the function $f(x, y) = x^3 + y^3 - 3x - 3y$f(x, y) = x3 + y3 − 3x − 3y is

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 4

*1 point*

*1 point*

For which of these functions are the mixed partial derivatives the same at $(0, 0)$(0, 0).

- ☐ $f(x, y) = 3x^2 + 7y^3 - 2x^2y^3$ f(x, y) = 3x2 + 7y3 − 2x2y3.
- ☐ $f(x, y) = \begin{cases} 1 & y \neq 0 \\ 0 & y = 0. \end{cases}$ f(x, y) = {10y = 0y = 0.
- ☐ $f(x, y) = |x|(y^2 + 1)$f(x, y) = |x|(y2 + 1).
- ☐ $f(x, y) = ax + by$f(x, y) = ax + by, where $a$a and $b$b are two fixed real numbers.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$f(x, y) = 3x^2 + 7y^3 - 2x^2 y^3$ f(x, y) = 3x2 + 7y3 − 2x2y3.
$f(x, y) = ax + by$ f(x, y) = ax + by, where $a$a and $b$b are two fixed real numbers.

Check Answers
Your score is: 0/6